AGENTIC QA HACKATHON | DELIVERABLE

Cost Breakdown Report

Token usage, estimated spend and cost-efficiency of the DemoBank QA Agent

Project	DemoBank QA Agent (Python / Playwright / Gemini)
Document	Cost Breakdown Report, version 1.1
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Data source	audit.json token counts; Google Gemini API pricing

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1. Executive Summary

1.1 Purpose

This report documents what it costs to run the DemoBank QA Agent: how many tokens were used, how the cost is calculated, and how it would scale across repeated runs. It accompanies the QA Audit Report as the cost disclosure for the Agentic QA Hackathon submission.

1.2 Key Figures

Metric	Value
LLM cost of the audit run (Gemini 2.5 Flash, current published rates)	$0.0135
Cost of the 17 browser checks	$0.0000 (deterministic Playwright, no LLM)
Tokens used (single LLM call)	1,263 input + 5,257 output = 6,520
Browser execution time	50.8 seconds
Cost per extracted requirement (125)	$0.000108

1.3 Main Conclusions

- **The audit is extremely inexpensive.** A complete run costs about $0.0135 (roughly 1.4 US cents), and all 17 test checks are free to execute.
- **The only paid component is requirement extraction** by Gemini 2.5 Flash. Output tokens account for about 97% of that cost.
- **Repeat runs can cost nearly nothing** if the extracted requirements are cached, since the PRD rarely changes (Section 8).

> **Basis of the figures:** costs are estimates computed from the recorded token counts at Google's published paid-tier list prices. They are not taken from a provider invoice. If the run used a free-tier API key, the amount actually billed would be $0.00.

2. Cost Model and Architecture

2.1 Where Costs Arise

The agent is deliberately hybrid: language-model capability is used only where it adds value (reading the PRD), while test execution is scripted. This keeps the recurring cost of testing at zero.

Component	Technology	Billable?	Run cost
PRD requirement extraction	Gemini 2.5 Flash API (prd_loader.py)	Yes, per token	$0.0135
Browser checks (17)	Playwright + Chromium (checks.py)	No	$0.0000
API balance assertions	Playwright request context with session cookie	No	$0.0000
Report generation	Python (reporter.py)	No	$0.0000

Component	Technology	Billable?	Run cost
Compute and hosting	Local machine; DemoBank run locally	Not recorded	Not included

2.2 Cost Flow

In `prd_loader.load_requirements()`, the PRD text is sent to Gemini 2.5 Flash with a prompt asking for a JSON array of requirements. The response's usage metadata (prompt and candidate token counts) is stored in the `llm_usage` dictionary. `runner._default_costs()` converts these counts to dollars and writes two `CostEntry` rows, which `reporter.py` prints in the Cost estimate section of audit.md and audit.json.

3. Cost Calculation

3.1 Token Usage and Cost Table

Label	Model	Input tokens	Output tokens	Est. USD
Automated browser checks	n/a (deterministic Playwright)	0	0	$0.0000
PRD Extraction	gemini-2.5-flash	1,263	5,257	$0.0135
Total		**1,263**	**5,257**	**$0.0135**

3.2 Calculation

Item	Calculation	Result
Input cost	1,263 / 1,000,000 x $0.30	$0.000379
Output cost	5,257 / 1,000,000 x $2.50	$0.013142
Total	Input + output	$0.013521

4. Pricing Basis

4.1 Published Rates

Costs use the Gemini 2.5 Flash paid-tier rates published by Google (checked on 20 October 2026).

Item	Paid tier, per 1M tokens (USD)
Input (text / image / video)	$0.30
Output, including thinking tokens	$2.50
Context caching (text / image / video)	$0.03

4.2 Thinking Tokens

Gemini 2.5 Flash is a hybrid reasoning model, and thinking tokens are billed at the output rate. The agent records the visible response token count. If thinking was active for this call, the true output total and cost would be somewhat higher. The size of this effect was not measured; the table below is illustrative only.

Hypothetical thinking tokens	Extra cost at $2.50 / 1M	Total cost
0 (as recorded)	$0.0000	$0.0135
2,000	$0.0050	$0.0185
5,000	$0.0125	$0.0260
10,000	$0.0250	$0.0385

4.3 Free Tier and Billing Context

Google offers a free tier for Gemini 2.5 Flash in which usage is free of charge. If the hackathon run used a free-tier key, the amount actually charged would be zero, and the figures in this report represent the notional cost of the same usage on the paid tier.

4.4 Sources

- Google AI for Developers, Gemini API pricing page (ai.google.dev/gemini-api/docs/pricing): Gemini 2.5 Flash paid-tier rates.
- SD Times, "Gemini 2.5 Pro and Flash are generally available and Gemini 2.5 Flash-Lite preview is announced": Flash-Lite rates used only for the illustrative comparison in Section 6.

5. Token Analysis

Metric	Value	Comment
Input tokens	1,263	Instruction prompt plus the PRD text.
Output tokens	5,257	JSON array of 125 requirement objects.
Total tokens	6,520	
Output share of tokens	80.6%	Generation dominates volume.
Output share of cost	97.2%	Output is priced about 8.3 times higher than input.
Output tokens per requirement	42.1	Average across the 125 extracted items.

The cost is driven almost entirely by output length. Any change that shortens the response (shorter field names, omitting the section text, or extracting only new requirements) lowers the cost proportionally; reducing the input prompt has a much smaller effect.

6. Unit Economics and Projections

6.1 Cost of One Extraction

Model (paid tier)	Input rate	Output rate	Cost per extraction
Gemini 2.5 Flash (used)	$0.30	$2.50	$0.0135
Gemini 2.5 Flash-Lite (illustrative comparison)	$0.10	$0.40	$0.0022

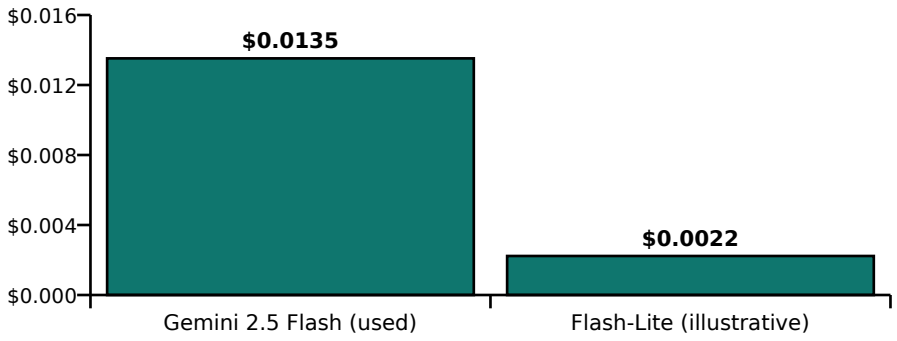


Figure 1. Estimated cost of one PRD extraction. Flash-Lite is shown for comparison only; its extraction quality was not tested.

6.2 Unit Costs

Unit	Cost
Per run (one extraction)	$0.013521
Per extracted requirement (125)	$0.000108
Per executed check (17)	$0.000795
Per confirmed application defect (3)	$0.004507

The cost per defect is illustrative: it divides the LLM spend by the three application defects confirmed in the QA Audit Report and excludes engineering time.

6.3 Projection Across Repeated Runs

Without caching, every run repeats the extraction. The table assumes each run uses the same token counts.

Number of runs	No caching	With caching (extract once)
1	$0.0135	$0.0135
4	$0.0541	$0.0135
10	$0.1352	$0.0135
100	$1.3521	$0.0135
1,000	$13.5214	$0.0135

The transaction history shows at least four runs on 15 October 2026 (at 5:51, 5:52, 6:26 and 6:31 PM PDT). If every one of those runs performed an extraction, the development-phase spend would be roughly four times the single-run figure, about $0.0541. Earlier runs were not logged, so this cannot be confirmed.

7. Time and Resource Cost

Resource	Measurement	Note
Browser execution time	50.8 s	From started_at to finished_at in audit.json.
LLM extraction latency	Not recorded	started_at is set after extraction, so this time is excluded.
Compute	Local machine, headless Chromium	No cloud compute cost; electricity and hardware not counted.
Network egress	Local loopback to DemoBank; one HTTPS call to Gemini	Negligible.
Human effort	Not included	Authoring checks, reviewing results and writing documentation.

8. Cost Optimisation Recommendations

Priority	Recommendation	Expected effect
1	Cache extracted requirements keyed by a hash of the PRD text, and call Gemini only when the PRD changes.	Repeat runs cost $0.00; removes the multiplier on repeated runs.
1	Read thoughts_token_count and add it to the output total, or set the thinking budget to 0 for this simple extraction task.	No hidden thinking cost; possibly a lower real cost.
2	Constrain the output schema (short keys, no restated section text) and reuse the stable REQ- ID scheme.	Fewer output tokens; also fixes the requirement ID mismatch.
3	Persist a cumulative cost ledger across runs (for example a cost.jsonl file).	Total project spend becomes auditable.
3	Evaluate a lighter model for extraction, measuring quality as well as price.	Up to about 80% lower extraction cost, if quality holds.

9. Assumptions and Limitations

- Token counts are those recorded in audit.json for the final run; earlier runs were not logged.
- Dollar figures are computed from those token counts at Google's published paid-tier prices as of 20 October 2026, which may change.
- Estimates exclude thinking tokens (not recorded), batch or caching discounts, taxes, and any free-tier allowance.
- Hosting, development time, the local machine, and the DemoBank deployment are outside the measured cost.
- Projections assume identical token counts per run; a longer PRD would increase both input and output tokens.

10. Conclusion

The DemoBank QA Agent delivers a full specification-driven audit for a marginal cost of about $0.0135 per run, with every browser check executing at no cost. Caching the extracted requirements and capturing thinking tokens would make the cost reporting more complete and bring the recurring cost of repeat audits to effectively zero.

Appendix A: Calculation Worksheet

Step	Formula	Value
Input tokens	From audit.json	1,263
Output tokens	From audit.json	5,257
Input cost	1,263 x 0.30 / 1,000,000	$0.000379
Output cost	5,257 x 2.50 / 1,000,000	$0.013142
Total LLM cost	Sum	$0.013521
Flash-Lite total (illustrative)	1,263 x 0.10 + 5,257 x 0.40, per 1,000,000	$0.002229
Run duration	finished_at minus started_at	50.83 s